11. Sequences and Series

11.4. Comparison Tests for Series

11.4 Learning Objectives
- Determine when it is appropriate to use a comparison test
- Determine an appropriate series to compare a given series to
- Know when and how to use direct comparison and limit comparison tests.

11.4.2. Limit Comparison Test

Theorem 11.4.1 The Limit Comparison Test

Suppose that $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are series with positive terms. If

$$\lim_{n\to\infty} \frac{a_n}{b_n} = c$$

where c is a finite positive number ($c > 0$), then either both seris converge, or both series diverge.

Example 11.4.1

Does the series $\sum_{n=1}^{\infty} \frac{1}{2^n - 1}$ converge or diverge?

Solution

Always checking the divergence test at the start, we can confirm the terms of the series go to zero, $\lim_{n\to\infty} \frac{1}{2^n-1} = 0$ so the divergence test is inconclusive.

Then, we can see that $\frac{1}{2^n-1}$ is fairly similar to $\frac{1}{2^n}$ which we know converges because $\sum_{n=1}^{\infty} \frac{1}{2^n} = \sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^n$ is a geometric series with $r = \frac{1}{2} < 1$.

Next, if we want to use direct comparison test to show convergence, then we would need

$$\frac{1}{2^n - 1} < \frac{1}{2^n}$$
$$\text{in other words} \quad 2^n - 1 > 2^n$$

which is false! So we **cannot** use direct comparison test.

We can use limit comparison test, however: let $b_n = \frac{1}{2^n}$ and $a_n = \frac{1}{2^n-1}$ then

$$\begin{aligned}\lim_{n\to\infty}\frac{a_n}{b_n} &= \lim_{n\to\infty}\frac{\frac{1}{2^n-1}}{\frac{1}{2^n}}\\ &= \lim_{n\to\infty}\frac{2^n}{2^n-1}\\ &= \lim_{n\to\infty}\frac{1}{1-\frac{1}{2^n}}\\ &= \frac{1}{1-0}\\ &= 1\end{aligned}$$

since 1 is constant, then **by the limit comparison test** the series $\sum_{n=1}^{\infty}\frac{1}{2^n-1}$ converges.

11.4b Section Summary: